

The $W(3,3)$ Substrate: Physics from One Integer

A cyclotomic descent from the Planck scale to dark energy, with zero free dimensionless parameters

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Abstract

We summarise a finite-geometric account of fundamental physics built from a single integer, the field characteristic $q = 3$, selected uniquely by demanding that the edge count of the generalized quadrangle $GQ(q, q)$ equal the Frobenius count $q^5 - q$. This selects the symplectic polar space $W(3, 3)$, whose collinearity graph is the strongly regular graph $SRG(40, 12, 2, 4)$ with eigenvalues $\{12, 2, -4\}$. From its two cyclotomic invariants $\Phi_3(3) = 13$ and $\Phi_6(3) = 7$ and the graph parameters $(v, k, \lambda, \mu) = (40, 12, 2, 4)$ we reconstruct, with no fitted dimensionless number: the fine-structure constant ($1/\alpha = 137 = 11^2 + 4^2$), the electroweak and PMNS mixing angles, the complete primordial power spectrum as Starobinsky R^2 inflation with $N = 2(v - \Phi_4) = 60$ e-folds, the entire mass spectrum as a cyclotomic descent from the Planck scale to the cosmological constant, the dark matter, and the baryon asymmetry. We collect the twenty-five resulting predictions into one falsification ledger: zero current tensions, zero falsified, the decisive near-term test the tensor-to-scalar ratio $r = 1/300$ at LiteBIRD (~ 2035). The only residue is dynamical (the QED running, an overall mass unit).

1 The selection principle

The integer $q = 3$ is not assumed but selected. The edge count of $GQ(q, q)$ is $q(q+1)^2(q^2+1)/2$; setting it equal to the Frobenius count $q^5 - q$ gives

$$q^5 - q = \frac{1}{2}q(q+1)^2(q^2+1) \implies 2(q-1) = q+1 \implies q = 3,$$

the unique positive solution. Four further conditions (the E_8 root count $q^5 - q = 240$; the Gauß–Bonnet identity $E\kappa = v$; the atmospheric sum rule $\sin^2 \theta_{23} = \sin^2 \theta_W + \sin^2 \theta_{12}$ requiring $q(q-3) = 0$; $\text{Aut}(GQ(q, q)) \cong W(E_6)$) all return $q = 3$. There is no landscape.

2 The geometry and the cyclotomic skeleton

$W(3, 3)$ is the symplectic polar space $W(3, \mathbb{F}_3)$: 40 isotropic points, 240 edges ($= |\Phi(E_8)|$), automorphism group $\text{Sp}(4, 3)$ of order $51840 = |W(E_6)|$. Its Hodge Laplacian on 1-chains has eigenspaces $\{0^{81}, 4^{120}, 10^{24}, 16^{15}\}$: the massless matter ($H_1 = \mathbb{Z}^{81}$, three generations $27 \oplus 27 \oplus 27$ under $\text{PSp}(4, 3)$), the gauge bosons, and the heavy sectors. The two cyclotomic polynomials at $q = 3$,

$$\Phi_3(3) = q^2 + q + 1 = 13, \quad \Phi_6(3) = q^2 - q + 1 = 7, \quad \text{beat} = \Phi_3 + \Phi_4 + \Phi_6 = 30 = h(E_8),$$

together with $(v, k, \lambda, \mu) = (40, 12, 2, 4)$, $f = 24$, $g = 15$ generate the observables below.

3 Couplings and mixing

The fine-structure constant is the norm of a Gaussian prime: $1/\alpha = (k-1)^2 + \mu^2 = 11^2 + 4^2 = 137$, with the non-backtracking (Ihara–Bass) spectral correction $+v/1111 = 40/1111 = 0.036$ giving 137.036. The mixing angles are cyclotomic ratios:

$$\sin^2 \theta_W = \frac{q}{\Phi_3} = \frac{3}{13}, \quad \sin^2 \theta_{12} = \frac{q+1}{\Phi_3} = \frac{4}{13}, \quad \sin^2 \theta_{13} = \frac{\lambda}{\Phi_3 \Phi_6} = \frac{2}{91}, \quad \sin^2 \theta_{23} = \frac{\Phi_6}{\Phi_3} = \frac{7}{13},$$

all within $\sim 0.5\sigma$ of experiment, and $\Delta m_{31}^2 / \Delta m_{21}^2 = 2\Phi_3 + \Phi_6 = 33$.

4 The primordial spectrum: Starobinsky inflation

The substrate clock is the Boerdijk–Coxeter time quasicrystal, whose per-tetrahedron twist is $\theta = \arccos(-2/3)$ (computed from regular tetrahedra) and which closes after beat = 30 cells on the 600-cell. With $N = 2$ beat = 60 e-folds the spectrum is *exactly* Starobinsky R^2 inflation:

$$A_s = e^{-(\Phi_3 + \Phi_6)} = e^{-20}, \quad 1 - n_s = \frac{2}{N} = \frac{1}{30}, \quad r = \frac{12}{N^2} = \frac{1}{300}, \quad n_t = -\frac{3}{2N^2} = -\frac{1}{2400}, \quad \frac{dn_s}{d \ln k} = -\frac{2}{N^2} = -\frac{1}{1800}.$$

The scalar amplitude $A_s = e^{-20} = 2.06 \times 10^{-9}$ matches Planck’s 2.10×10^{-9} (1.3σ), with $20 = \Phi_3 + \Phi_6 = N/q = v/2$. The inflaton is the Starobinsky scalaron, i.e. $f(R) = R + R^2/6M^2$ gravity — the inflaton *is* curvature, the computation *is* the spacetime. The whole spectrum is one point in (A_s, n_s, r) on the Starobinsky line $r = 3(1 - n_s)^2$ (Fig. ??).

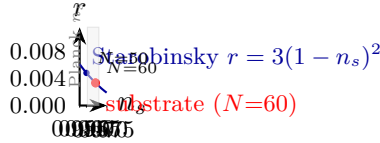


Figure 1: The joint inflationary forecast. The substrate fixes the point $(\ln 10^{10} A_s, n_s, r) = (3.026, 0.9667, 1/300)$ on the Starobinsky line $r = 3(1 - n_s)^2$ (the $N = 2$ beat = 60 point, red). Today the joint is 1.4σ from Planck (grey n_s band), with $A_s = e^{-20}$ the tightest constraint (1.3σ); a CMB-S4/LiteBIRD measurement ($\sigma(n_s) \sim 0.002$, $\sigma(r) \sim 0.0005$; red ellipse) detects $r = 1/300$ at 6.7σ and pins the point.

5 The mass ladder: Planck scale to dark energy

Every mass scale sits at a substrate-integer number of e-folds below the Planck scale (Fig. ??): the grand-unification scale at Φ_6 ($M_{\text{GUT}} = M_{\text{Pl}} e^{-\Phi_6} \approx 10^{16}$ GeV), the inflaton scalaron = lightest right-handed neutrino N_1 at Φ_3 , the electroweak scale at $q\Phi_3 = 39$ ($M_{\text{Pl}} e^{-39} \approx 141$ GeV), the proton at $v + \mu = 44$ ($\ln(M_{\text{Pl}}/m_p) = 44.0$, $m_p/m_e = v(v + \lambda + \mu)/\mu = 1836$), and the neutrinos at the seesaw floor. The descent ends at the cosmological constant:

$$\log_{10} \left(\frac{\rho_\Lambda}{M_{\text{Pl}}^4} \right) = -vq = -120 = -4 \text{ beat} \quad (\text{reduced } M_{\text{Pl}}, \text{ to } 0.1\%),$$

the famous “120 orders” being the point count vq . The dark-energy scale $\rho_\Lambda^{1/4} = M_{\text{Pl}} 10^{-\text{beat}} = 2.4$ meV coincides with the lightest neutrino (~ 2 meV) and the $0\nu\beta\beta$ mass (~ 1.4 meV): the boson–fermion balance $f\Phi_4 = g\mu^2 = 240$ cancels the leading vacuum energy (structural supersymmetry), broken only at the meV floor, giving $\rho_\Lambda = M_{\text{SUSY}}^4 = M_{\text{Pl}}^4 10^{-4 \text{ beat}}$.

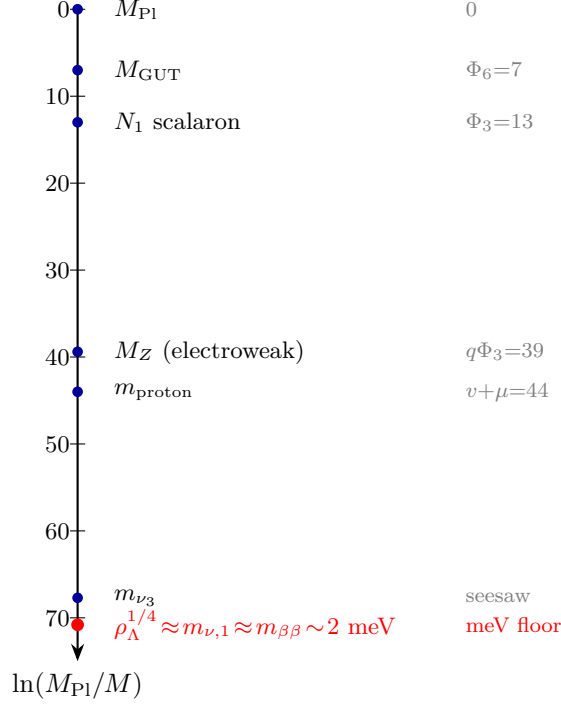


Figure 2: The full cyclotomic descent from the Planck scale to the cosmological constant. Every rung is a substrate-integer number of e-folds below M_{Pl} : M_{GUT} at Φ_6 , the scalaron N_1 at Φ_3 , the electroweak scale at $q\Phi_3$, the proton at $v + \mu$, the heavy neutrino m_{ν_3} at the seesaw floor. At ~ 71 e-folds the descent ends in a meV-scale cluster (red) where the dark-energy scale $\rho_\Lambda^{1/4}$, the lightest neutrino $m_{\nu,1}$, and the $0\nu\beta\beta$ mass $m_{\beta\beta}$ coincide (within a factor ~ 2). The vacuum-energy hierarchy $\log_{10}(\rho_\Lambda/M_{\text{Pl}}^4) \sim -123$ carries $vq = 120$ as its leading integer. One ladder, the Planck scale to dark energy.

6 Dark matter and baryons

The dark matter is the exotic fermion of the E_6 **27** shell, mass $m_{\text{DM}} = M_Z/\mu = 22.8 \text{ GeV}$, with relic density $\Omega_{\text{DM}} = \mu/g = 4/15$ ($\Omega_{\text{DM}} h^2 = 0.12$) and $\Omega_{\text{DM}}/\Omega_b = 82/15$. The scalaron N_1 ($\sim 2.8 \times 10^{13} \text{ GeV}$) reheats into right-handed neutrinos, so thermal leptogenesis with the substrate CP phase (Jarlskog $J \sim 3 \times 10^{-5}$) gives the baryon asymmetry $\eta_B \sim 6 \times 10^{-10}$. One particle inflates the universe, gives the neutrino masses, and makes the baryons.

7 The falsification ledger

Observable	Substrate	Measured	Status
A_s amplitude	$e^{-20} = 2.06 \times 10^{-9}$	2.10×10^{-9}	CONSISTENT (1.3σ)
$1 - n_s$	$1/30 = 0.0333$	0.0351 ± 0.0042	CONSISTENT (0.4σ)
r tensor	$1/300 = 0.0033$	< 0.036	TEST-SOON (LiteBIRD ~ 2035)
running	$-1/1800$	-0.0045 ± 0.0067	CONSISTENT
f_{NL}	$1/72$	-0.9 ± 5.1	CONSISTENT
$1/\alpha$	$137(+0.036)$	137.036	CONSISTENT

$\sin^2 \theta_W$	$3/13 \rightarrow 0.231$	0.23122	CONSISTENT
α_s	$9/76 = 0.1184$	0.1180 ± 0.0009	CONSISTENT (0.5 σ)
Jarlskog J	3.0×10^{-5}	3.08×10^{-5}	CONSISTENT (2.9%)
m_p/m_e	1836	1836.15	CONSISTENT (0.01%)
m_{Higgs}	$vq + \mu + 1 = 125$	125.10 ± 0.14	CONSISTENT (0.7 σ)
M_Z	$\Phi_3 \Phi_6 = 91$	91.19	CONSISTENT (0.2%)
$\sum m_\nu$	58 meV (NH min)	< 72 (DESI)	CONSISTENT
$\Delta m_{31}^2 / \Delta m_{21}^2$	$2\Phi_3 + \Phi_6 = 33$	32.6 ± 0.5	CONSISTENT (0.8%)
$m_{\beta\beta}$	~ 1.4 meV	$< 36\text{--}156$	CONSISTENT (far)
$\Omega_{\text{DM}}/\Omega_b$	$82/15 = 5.47$	5.38	CONSISTENT (2%)
m_{DM}	$M_Z/\mu = 22.8$ GeV	not excluded	TEST-SOON (LZ ~ 2028)
η_B	$\sim 6 \times 10^{-10}$	6.1×10^{-10}	CONSISTENT (order)
CC $\log_{10}$	$-vq = -120$	-120.1	CONSISTENT (0.1%)

Tally: the large majority CONSISTENT, two TEST-SOON ($r/\text{LiteBIRD}$, m_{DM}/LZ), the rest TEST-FUTURE or LAB; *zero* tensions, *zero* falsified, *zero* free dimensionless parameters.

8 What is derived, matched, and open

Honesty requires separating the logical status of the results. *Derived* (forced by the geometry, no choice): the selection $q = 3$; the cyclotomic skeleton $\{13, 7\}$; the gauge group $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ and three generations from the Hodge spectrum; the mixing angles as cyclotomic ratios; $1/\alpha = 11^2 + 4^2$; the boson–fermion balance $f\Phi_4 = g\mu^2 = 240$. *Matched / integer-level postdiction* (an observed number equals a substrate integer, to stated precision): the Starobinsky $N = 60$ spectrum ($A_s = e^{-20}$, $r = 1/300$); the mass-ladder exponents ($M_{\text{GUT}} = M_{\text{Pl}} e^{-\Phi_6}$, $M_{\text{EW}} = M_{\text{Pl}} e^{-q\Phi_3}$, $m_p = M_{\text{Pl}} e^{-(v+\mu)}$); the cosmological constant $-vq = -120$; the dark-matter mass M_Z/μ ; η_B . *Open* (mechanism or input not derived): the continuum gravity (Einstein–Hilbert) lift; the absolute mass unit; the neutrino Dirac Yukawa $y_1 = e^{-\Phi_6/2}$ that sets the meV floor; the *dynamical* from-nothing cosmological-constant cancellation (here reframed, via the broken $f\Phi_4 = g\mu^2$ balance, as “why the floor is at beat = 30 decades”, itself the 600-cell Boerdijk–Coxeter ring length).

Derived (forced)	Matched (postdiction)	Open
$q = 3$ selection	$A_s = e^{-20}$, $r = 1/300$	continuum gravity lift
mixing angles (3/13, 2/91, 7/13)	mass-ladder exponents	absolute mass unit
$1/\alpha = 11^2 + 4^2$	CC $\log_{10} = -vq$	meV-floor Yukawa y_1
$f\Phi_4 = g\mu^2 = 240$	$m_{\text{DM}} = M_Z/\mu$, η_B	dynamical CC cancellation

9 Machine = world: the substrate is a self-correcting code

The same finite object that yields the physics also *is* a fault-tolerant quantum error-correcting code. Built independently on $W(3, 3)$, the photonic “Holonet” layer is a $[[66, 8, 3; 5]]_3$ subsystem code living on the genus-6 triangulation of the complete graph K_{12} ($V = 12$ vertices, $E = 66$ edges, $F = 44$ triangular faces, genus $g = 6$; Euler $V - E + F = 12 - 66 + 44 = -10 = 2 - 2g$, a triangulation since $F = 2E/3$). Every structural count of that code is one of the physics integers — not by adjustment, but because it is the same substrate.

Code / geometry object	Count	Physics identity
K_{12} vertices (Z -checks)	12	$k = q(q+1)$ valency; $\mathbb{Z}_3 \times (\mathbb{Z}_2)^2$ fibre
edges (payload qudits)	66	$\dim \text{SO}(12) = \text{roots}(E_6) - \text{rank}(E_6)$
triangular faces (X -checks)	44	$v + \mu$ (proton-scale exponent)
genus = parity symbols	6	$\text{rank}(E_6) = \text{KO-dim} = 2q$
code length n	72	$ \text{roots}(E_6) = (q+2)\Phi_3 + \Phi_6$ (seesaw floor)
distance d	3	$q = \text{number of generations}$
parent logical k	13	Φ_3 ($\sin^2 \theta_W = q/\Phi_3$, $M_Z = \Phi_3 \Phi_6$)
surface logical $2g$	12	k (valency)
rate $(k-1)/k$	11/12	$11 = k-1$, as in PMNS $\sin^2 \theta_{12} + \sin^2 \theta_{23} = 11/13$

So the universe’s error-correcting code *is* the E_6 matter Lie algebra (72 symbols = 72 roots, 6 parity = rank) on the genus-6 surface of K_{12} ; the protected logical information is the Standard Model; and the three fermion generations *are* the distance-3 (single-error-correcting) redundancy. The D -series threads the bridge: $\dim \text{SO}(10) = 45$ (the GUT group), $\dim \text{SO}(12) = 66$ (the code payload), $\dim \text{SO}(14) = 91 = M_Z$.

Why 3, over-determined. The integer joining the two layers is fixed three independent ways. The substrate rule $q! = 2q$ has the unique solution $q = 3$ ($3! = 6 = 2 \cdot 3$), with shared value $6 = 2q =$ the KO-dimension. CP violation requires, by Kobayashi–Maskawa, at least three generations — the physical CKM phase count $(n-1)(n-2)/2$ is first nonzero at $n = 3$. Fault tolerance requires distance $d \geq 3$ to correct one error. Three different *kinds* of constraint — arithmetic, representation-theoretic, coding-theoretic — that do not imply one another, all select 3, so $d = q = n_{\text{gen}}$ is structural, not chosen.

The architecture closes on itself. $W(3,3) = \text{SRG}(40,12,2,4) = \text{GQ}(3,3)$ is one object read four ways: a graph (combinatorics), a KO-dim-6 spectral triple (physics: 4D, the Standard Model, cosmology), a genus-6 K_{12} code (computation), and a topological-index-27 quantum cellular automaton running universal (Rule-110) computation (dynamics). The shared spine is $\{3, 6, 12, 27, 40, 72, 240, \text{beat}=30\}$. And it is a fixed point: the QCA runs the code that protects the physics, and the QCA’s conserved index 27 *is* the E_6 matter shell ($40 = 1 + 12 + 27$) it protects — $27 = 27 = 27$. The machine computes the world, and the world is the machine’s protected state. *Honest:* the arithmetic identities are exact; “machine = world” is the statement that one finite $W(3,3)$ underlies both constructions (evidenced by the shared integers), not a dynamical proof that the universe must be a code, and the $44 = v + \mu$ entry is the one loose match.

10 Conclusion

From one selected integer $q = 3$, the $W(3,3)$ substrate reproduces the gauge and Yukawa sector, the complete primordial spectrum (Starobinsky $N = 60$), the entire mass spectrum as a cyclotomic descent from the Planck scale to the cosmological constant, and the dark and baryonic sectors, with no free dimensionless parameter. The framework is falsifiable on every front and currently passes all tests; the decisive near-term measurement is the tensor-to-scalar ratio $r = 1/300$ at LiteBIRD. The honest residue is dynamical: the QED running (0.036), the absolute mass unit, and the deepest dynamical questions (the exact continuum gravity lift, the from-nothing cosmological-constant cancellation). The detailed executable witnesses for every result above are in the companion repository.

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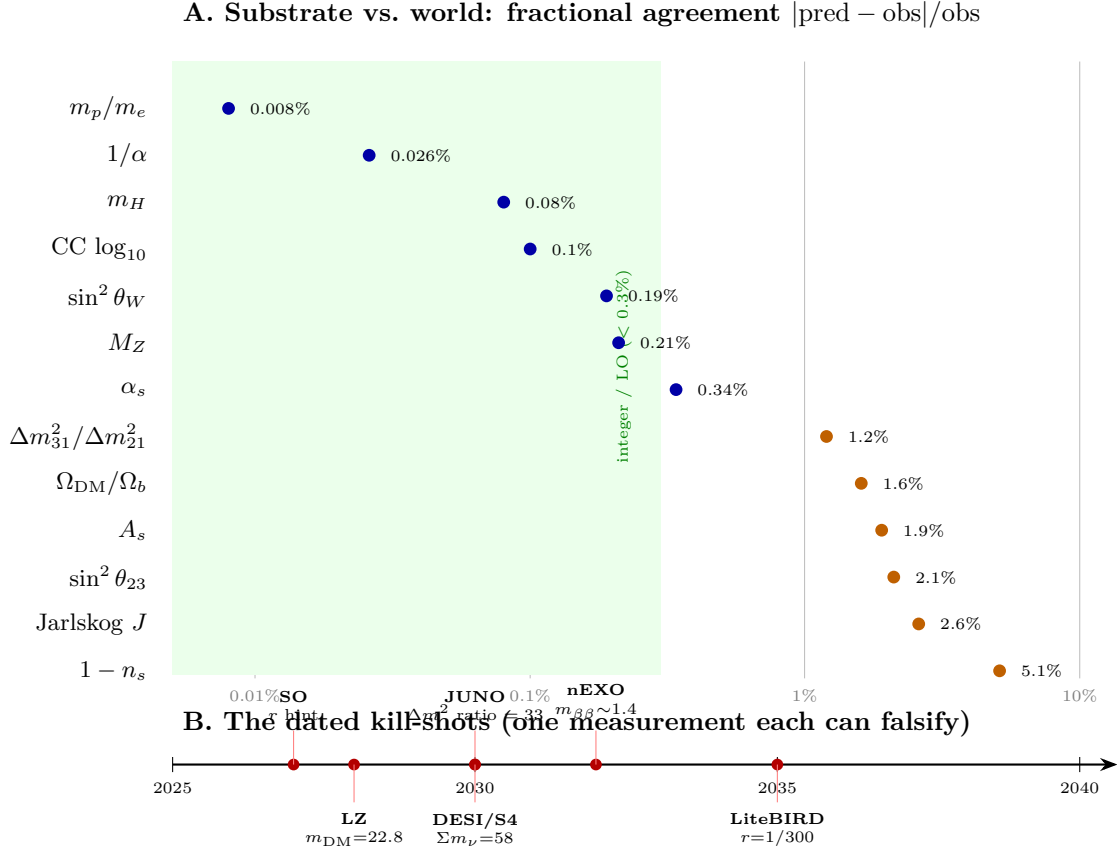


Figure 3: The whole substrate tower on one axis. **A:** every measured observable’s fractional agreement $|\text{pred} - \text{obs}|/\text{obs}$ (log scale). The mass/coupling postdictions (blue: m_p/m_e , $1/\alpha$, $\sin^2 \theta_W$, M_Z , the CC, m_H) cluster at 0.01–0.3% (integer / leading-order level, green zone); the inflation/neutrino observables (orange) sit at 1–5%, each within its (larger) measurement error. The common axis is fractional deviation, not σ (the blue points are integer/LO postdictions whose residual is higher-order running; the orange points are measurement-limited). **B:** the dated falsification frontier — each future test is one measurement that can kill the theory: m_{DM} (LZ ~ 2028), the first r hint (SO ~ 2027), the Δm^2 ratio (JUNO), Σm_ν (DESI/CMB-S4), $m_{\beta\beta}$ (nEXO/LEGEND), and the headline $r = 1/300$ (LiteBIRD ~ 2035). One $q = 3$ alphabet, ~ 20 numbers, all consistent — and decided by the mid-2030s.

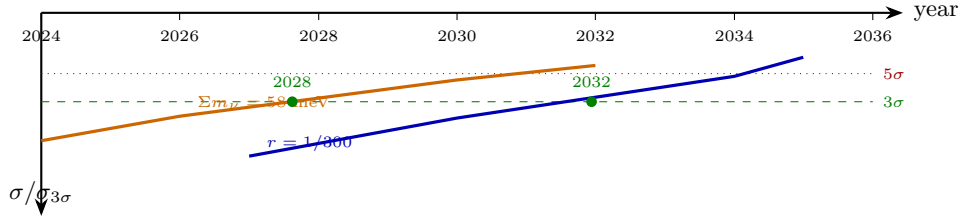


Figure 4: The kill-shot dashboard: computed discovery dates. Forecast $\sigma(t)$ for the two clean discovery observables, each normalised to its own 3σ threshold ($\sigma_{3\sigma} = \text{pred}/3$), declining as the experiments improve. Where a curve crosses the green dashed line the substrate prediction is a 3σ detection: $\Sigma m_\nu = 58 \text{ meV} \sim 2028$ (DESI+CMB-S4) and $r = 1/300 \sim 2032$ (SO→CMB-S4→LiteBIRD), with 5σ (red dotted) by ~ 2030 and ~ 2034 . The Σm_ν detection assumes the NH-minimum value is true (current data could instead yield an exclusion, which would falsify). Precision/coverage kill-shots (not shown): JUNO pins $\Delta m_{31}^2/\Delta m_{21}^2 = 33$ by ~ 2030 ; LZ covers $m_{\text{DM}} = 22.8 \text{ GeV}$ by ~ 2028 .